MATH208
Data Structure

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Lectures

- Class hours: Mon. (9:10-12:00)
  - Classroom: 理 SC 2004
- Lecture: Szu-Chi Chung (鍾思齊)
  - Office: 理 SC 2002-4
  - Office hours: Mon. 16:00~18:00 and Wed. 16:00~18:00
- T.A.:黃冠欽
  - Office: SC2003-2
  - TA hour: Wed. 16:00~17:00

Textbook and requirement

- The assignment and related material will be available on the course webpage. Course website and Facebook group
  - https://phonchi.github.io/nsysu-math208/
  - https://www.facebook.com/groups/270467082661736 5
  - https://phonchi.github.io/ds-cpp-selfstudy/
- Textbook: *Problem Solving with Algorithms and Data Structures using C++*
  - Authors: Brad Miller, David Ranum and Jan Pearce
- Problem Solving with Algorithms and Data Structures using Python, 3rd Edition
  - Authors: Brad Miller and David Ranum
- For the basic data structure code checkout
  - https://github.com/phonchi/pythonds3/tree/master/cppds

Grading policy

- Grading
  - Homework 50% (5 assignments, mainly coding parts (C++/Python)
  - Participants: 5% (participates at least 10 times can get the full score)
  - Take-home Quiz: 5% (2 times, conceptual, single choice and short answer questions)
  - Midterm exam 20% (conceptual, single choice and short answer questions)
  - Final exam 20% (conceptual, single choice and short answer questions)
- Midterm (conceptual)
  - It will be held on 2026/11/02 at 理 SC1001
- Final (conceptual):
  - It will be held on 2026/12/21 at 理 SC1001
- You need to use C++/Python in this course

Resources

- C++/ Python basics
  - https://ocw.aca.ntu.edu.tw/ntu-ocw/ocw/cou/111S107 or Python
- Object-oriented
  - https://openhome.cc/Gossip/CppGossip/ or https://openhome.cc/zh-tw/python/
- Practicing
  - Hackerrank
  - https://neetcode.io/practice
- Visualization
  - C++/Python Tutor
  - https://visualgo.net/en
  - https://opendsa-server.cs.vt.edu/home/about
  - https://cmps-people.ok.ubc.ca/ylucet/DS/Algorithms.html

What we are going to study in this semester

- C++/ Python Object oriented fundamentals
  - Variables and containers
  - Flow Control
  - Functions
  - Encapsulation, polymorphism and inheritance (Classes)
- Algorithm analysis
  - Big O
  - Timing benchmark
- Linear structure
  - Array/Linked list
  - Stack/Queue
- Recursion
- Searching and Sorting
- Nonlinear data structure
  - Hash tables
  - Graphs
  - Trees
    - Binary search tree
    - AVL tree
- Not covered
  - Advanced graph and dynamic programming
  - Divide and conquer and Greedy algorithm
  - Network flow
  - NP and computational intractability

Relate to other courses

- Related Courses
  - Discrete Mathematics
  - Algorithms
  - Python and machine learning algorithms
- Other courses
  - Advance programming
  - Web programming
  - Network programming
  - Software engineering
  - Data science/Machine learning/Artificial intelligence

Some notes on using AI

- The Shift from Determinism to Non-determinism
  - AI models work based on probabilities, meaning the same prompt might yield different results. AI can provide code that works in one scenario but fails in another
- The Role of AI in Development
  - AI reduces the manual labor of typing syntax, allowing developers to focus more on high-level logic and problem-solving
  - AI should be treated as a fast but sometimes unreliable intern, so human verification is mandatory
  - Use AI for explaining complex code, generating unit tests, or suggesting alternative solutions where understanding context is key
- Impact on Learning: Foundations Still Matter
  - If beginners jump straight to "Vibe Coding" without understanding logic, they risk becoming "Prompt Engineers" who cannot fix systems when AI fails